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SIR Epidemic on a Graph

Simon Frost 2026-05-14

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Introduction

In contrast to the edge-based approach of [EdgeBasedModels.jl](#), which works with degree distributions and probability generating functions, `NodeBasedModels.jl` operates directly on specific graph instances. This means every node and every edge is tracked explicitly, allowing us to capture the full heterogeneity of a network — not just its degree distribution, but the precise adjacency structure.

[Sharkey \(2011\)](#) showed that node-level epidemic models can be organised into a hierarchy of moment closures:

- Order-1 (individual-based / NIMFA): Tracks the marginal probability $\langle S_i \rangle, \langle I_i \rangle$ for each node i . Pairs are approximated by independence:

$$\langle S_i I_j \rangle \approx \langle S_i \rangle \langle I_j \rangle.$$

- Order-2 (pair-based): Tracks both node marginals and pair probabilities $\langle S_i I_j \rangle$ for each edge (i, j) . Triples are closed via the Kirkwood superposition:

$$\langle A_k B_i C_j \rangle \approx \frac{\langle A_k B_i \rangle \langle B_i C_j \rangle}{\langle B_i \rangle}.$$

- Exact (Gillespie): Continuous-time Markov chain simulation using the Gillespie algorithm. No approximation — the “gold standard.”

Higher-order closures are more accurate but more expensive. In this vignette we run all three levels on the same graph and compare their predictions for an SIR epidemic.

Setup

```
using NodeBasedModels
using Graphs
using Plots
using OrdinaryDiffEqDefault
using Random
```

Create a graph

We build a random regular graph with $N = 100$ nodes and degree $k = 6$. Every node has exactly 6 neighbours.

[!NOTE]

$R_0 = 2$ anchor. For the homogeneous pairwise anchor, $R_0 = \tau(k-2)/\gamma$. With $k = 6$ and $\gamma = 0.25$, the comparable per-edge/pair rate is $\tau = 0.125$, and 1% seeding corresponds to one initial node.

```
Random.seed!(42)
N = 100
k = 6
γ_val = 0.25
R0_target = 2.0
τ_val = R0_target * γ_val / (k - 2)
initial_seed = [1]
tmax = 40.0
g = random_regular_graph(N, k)
net = GraphNetwork(g)
```

```
println("Nodes:      ", nv(g))
println("Edges:      ", ne(g))
println("Mean degree: ", mean_degree(net))
```

```
Nodes:      100
Edges:      300
Mean degree: 6.0
```

Individual-based model (order-1)

The individual-based (NIMFA) approximation tracks the probability that each node is in a given state. For the SIR model on a graph with adjacency matrix A , the equations for node i are

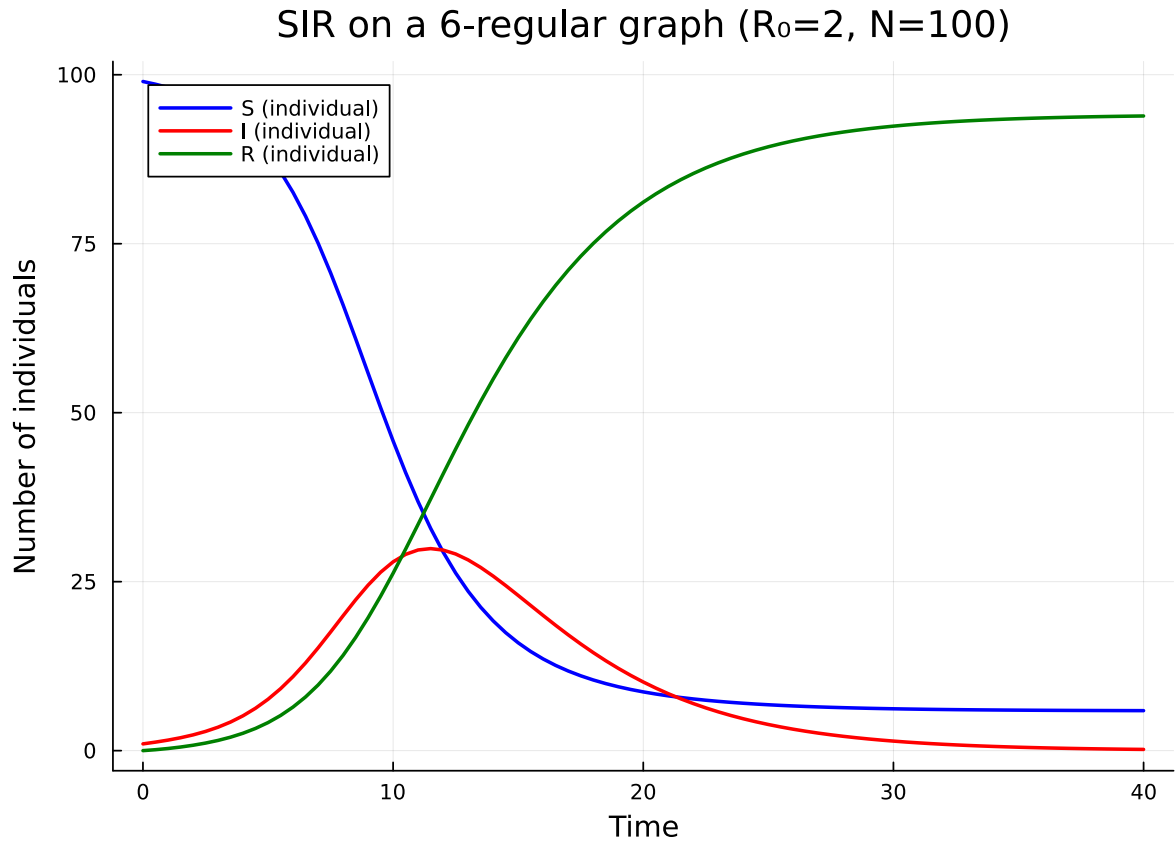
$$\frac{d\langle S_i \rangle}{dt} = -\tau \sum_j A_{ij} \langle S_i \rangle \langle I_j \rangle, \quad \frac{d\langle I_i \rangle}{dt} = \tau \sum_j A_{ij} \langle S_i \rangle \langle I_j \rangle - \gamma \langle I_i \rangle.$$

Because pairs are factorised ($\langle S_i I_j \rangle \approx \langle S_i \rangle \langle I_j \rangle$), the system has $2N$ ODEs.

```
ib_result = generate_individual_based(
    sir_model(), net;
    infection_rate = tau_val,
    recovery_rate  = gamma_val,
    initial_infected = initial_seed,
    tspan = (0.0, tmax),
    saveat = 0.5
)

S_ib = aggregate(ib_result, :S)
I_ib = aggregate(ib_result, :I)
R_ib = 100.0 .- S_ib .- I_ib
t_ib = range(0.0, tmax, length = length(S_ib))

p = plot(t_ib, S_ib, label = "S (individual)", lw = 2, color = :blue,
        xlabel = "Time", ylabel = "Number of individuals",
        title = "SIR on a 6-regular graph (R_o=2, N=100)")
plot!(p, t_ib, I_ib, label = "I (individual)", lw = 2, color = :red)
plot!(p, t_ib, R_ib, label = "R (individual)", lw = 2, color = :green)
p
```



Pair-based model (order-2)

The pair-based model additionally tracks the joint probability $\langle S_i I_j \rangle$ for every directed edge (i, j) . Triples like $\langle S_k S_i I_j \rangle$ that appear in the pair equations are closed using the Kirkwood closure:

$$\langle A_k B_i C_j \rangle \approx \frac{\langle A_k B_i \rangle \langle B_i C_j \rangle}{\langle B_i \rangle}.$$

This closure is exact on tree graphs (no cycles), because in that case nodes k and j are conditionally independent given i .

```
pb_result = generate_pair_based(
    sir_model(), net;
    infection_rate = tau_val,
    recovery_rate = gamma_val,
    initial_infected = initial_seed,
    tspan = (0.0, tmax),
```

```

    saveat = 0.5
)

S_pb = aggregate(pb_result, :S)
I_pb = aggregate(pb_result, :I)
R_pb = 100.0 .- S_pb .- I_pb
t_pb = range(0.0, tmax, length = length(S_pb))

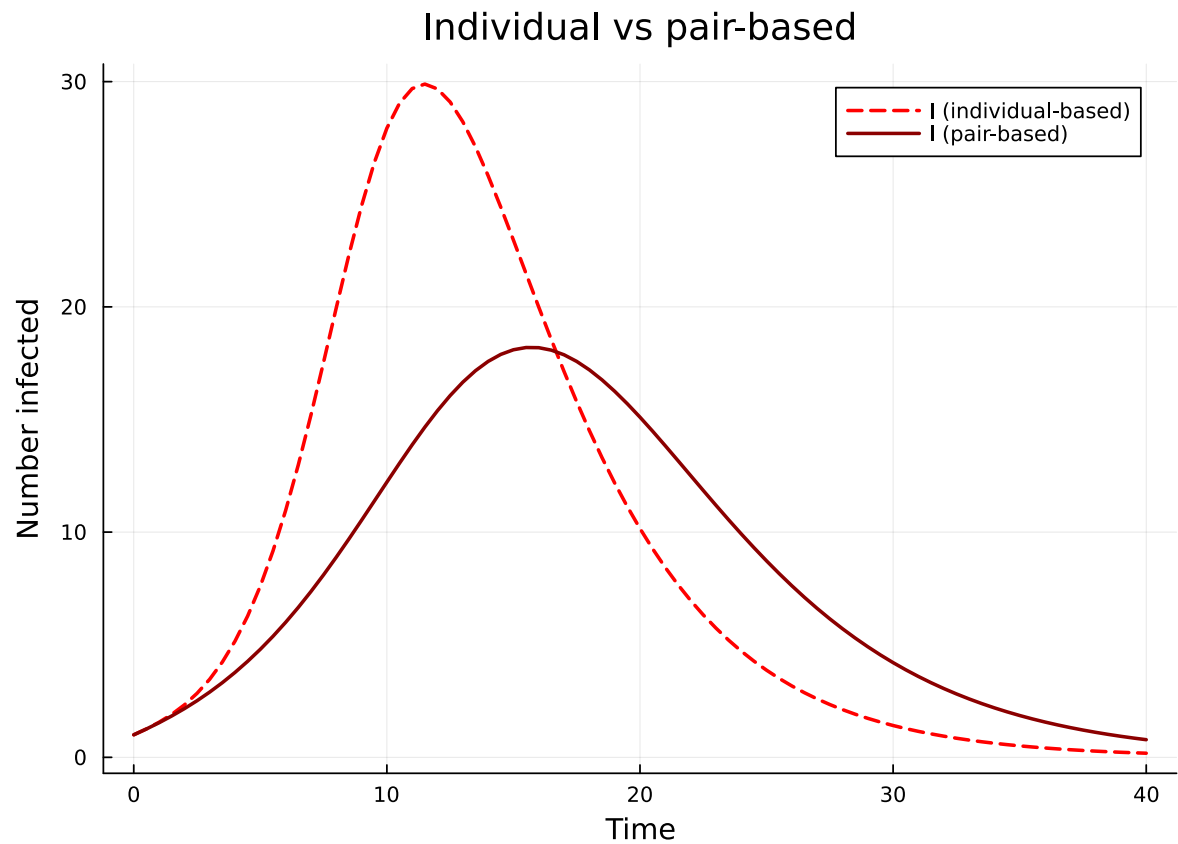
```

```
0.0:0.5:40.0
```

```

p = plot(t_ib, I_ib, label = "I (individual-based)", lw = 2, ls = :dash, color = :red,
        xlabel = "Time", ylabel = "Number infected",
        title = "Individual vs pair-based")
plot!(p, t_pb, I_pb, label = "I (pair-based)", lw = 2, color = :darkred)
p

```



Gillespie validation

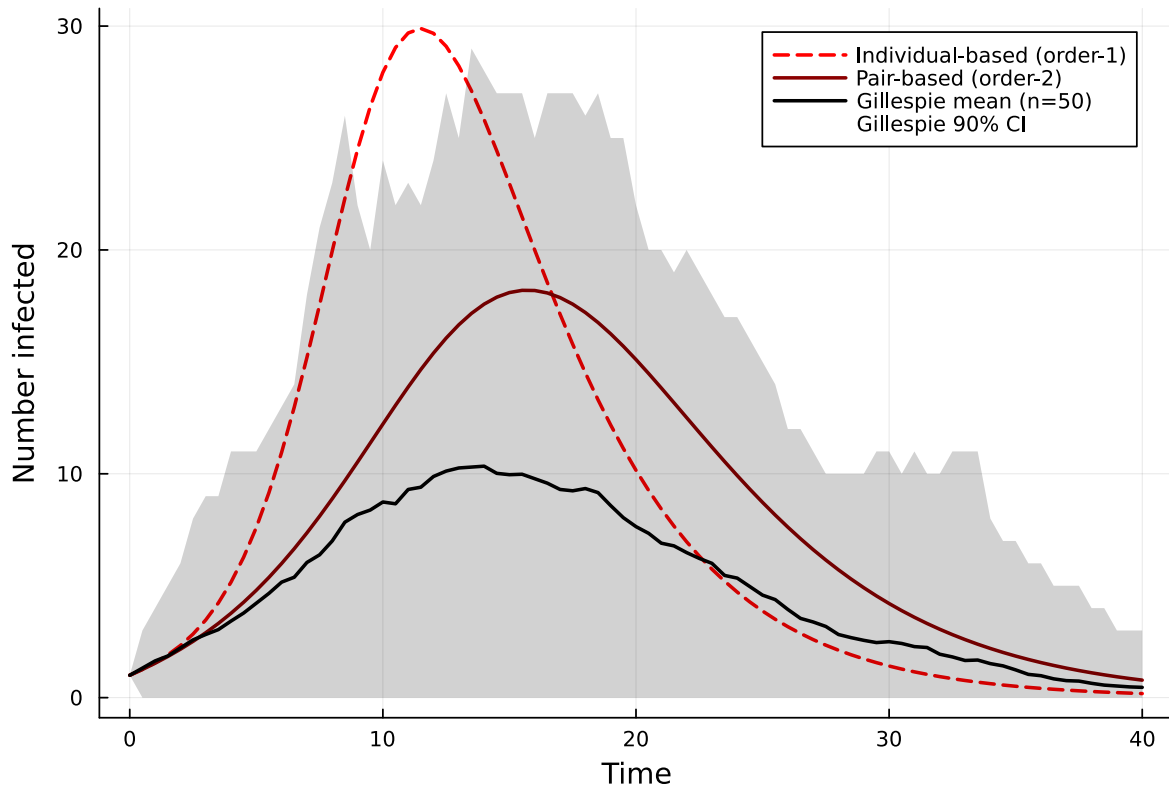
The Gillespie algorithm simulates the exact continuous-time Markov chain. We run 50 realisations and compute the mean trajectory together with a 90% confidence interval (5th–95th percentile envelope).

```
gill_avg = gillespie_sir_average(  
    net;  
    nruns          = 50,  
    dt             = 0.5,  
    tmax_grid      = tmax,  
    infection_rate =  $\tau_{\text{val}}$ ,  
    recovery_rate  =  $\gamma_{\text{val}}$ ,  
    initial_infected = initial_seed  
)
```

```
(t_grid = [0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5 ... 35.5, 36.0, 36.5, 37.0, 37.5, 38.0, 38.5, 39.0, 39.5, 40.0, 40.5, 41.0, 41.5, 42.0, 42.5, 43.0, 43.5, 44.0, 44.5, 45.0, 45.5, 46.0, 46.5, 47.0, 47.5, 48.0, 48.5, 49.0, 49.5, 50.0])
```

```
p = plot(t_ib, I_ib, label = "Individual-based (order-1)", lw = 2, ls = :dash, color = :red,  
        xlabel = "Time", ylabel = "Number infected",  
        title = "Moment closure hierarchy – SIR on 6-regular graph")  
plot!(p, t_pb, I_pb, label = "Pair-based (order-2)", lw = 2, color = :darkred)  
plot!(p, gill_avg.t_grid, gill_avg.I_mean, label = "Gillespie mean (n=50)",  
      lw = 2, color = :black)  
plot!(p, gill_avg.t_grid, gill_avg.I_mean,  
      ribbon = (gill_avg.I_mean .- gill_avg.I_q05, gill_avg.I_q95 .- gill_avg.I_mean),  
      fillalpha = 0.18, linealpha = 0.0, color = :black, label = "Gillespie 90% CI")  
p
```

Moment closure hierarchy — SIR on 6-regular graph



The plot illustrates the characteristic ordering of the hierarchy:

$$\text{Individual-based} \geq \text{Pair-based} \geq \text{Gillespie mean}.$$

The individual-based model over-estimates the epidemic because the pairwise independence assumption ignores dynamical correlations: if node i infected node j , then j is less likely to re-infect i because i is probably still infected. This “2-cycle” effect is captured by the pair-based model.

Discussion

NetworkOutbreaks ribbon

To standardise across the package suite we also overlay the [NetworkOutbreaks.jl](#) Gillespie SSA (used as the canonical ground truth in EBM and NO vignettes). At $N = 100$ this is on the small side for a smooth ribbon — we average across several graph realisations to reduce conditioning on a single graph.

```
include("../_validation.jl")

t_no, μ_no, σ_no = gillespie_ribbon(
    sir_model(τ = :β), # NBM model → NO adapter
    Dict{:β ⇒ τ_val, :γ ⇒ γ_val},
    regular_graph_builder(N, k);
    N = N, n_graphs = 5, nsims_per_graph = 20,
    tspan = (0.0, tmax), seed_fraction = length(initial_seed) / N)
```

```
([0.0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5 ... 35.5, 36.0, 36.5, 37.0, 37.5, 38.0,
```

```
plot(t_no, μ_no[:I] .* N, ribbon = σ_no[:I] .* N,
     label = "NO SSA (mean ± 1σ)", color = :darkred, fillalpha = 0.18, linealpha = 0.5)
plot!(t_ib, I_ib, label = "Order-1 individual-based",
      lw = 2, ls = :dash, color = :red)
plot!(t_pb, I_pb, label = "Order-2 pair-based",
      lw = 2, color = :darkred)
xlabel!("Time"); ylabel!("Number infected")
title!("NodeBasedModels closures vs NetworkOutbreaks SSA")
```

The pair-based closure tracks the SSA ribbon mean closely; the individual- based closure systematically over-predicts the peak, illustrating the moment-closure ordering numerically.

The three levels of the moment closure hierarchy trade off accuracy against computational cost:

Level	Variables	Closure	Exact when?
Order-1 (individual)	$O(N)$	Independence	Complete graph (mean-field limit)
Order-2 (pair-based)	$O(N + M)$	Kirkwood	Tree graphs (no cycles)
Exact (Gillespie)	$O(N)$ state	None	Always (stochastic)

The order-1 approximation ignores correlations introduced by 2-cycles (anomalous terms of the form $\langle S_i I_j \rangle - \langle S_i \rangle \langle I_j \rangle$). These terms are always positive during an epidemic, so order-1 systematically over-estimates the force of infection.

The order-2 approximation captures pair correlations but neglects triangles (3-cycles). On graphs with low clustering (like random regular graphs), the pair-based model is already very accurate. On highly clustered graphs, higher-order closures or the exact Gillespie simulation may be needed.

In the next vignette, we explore how different network topologies affect both the epidemic dynamics and the accuracy of these approximations.

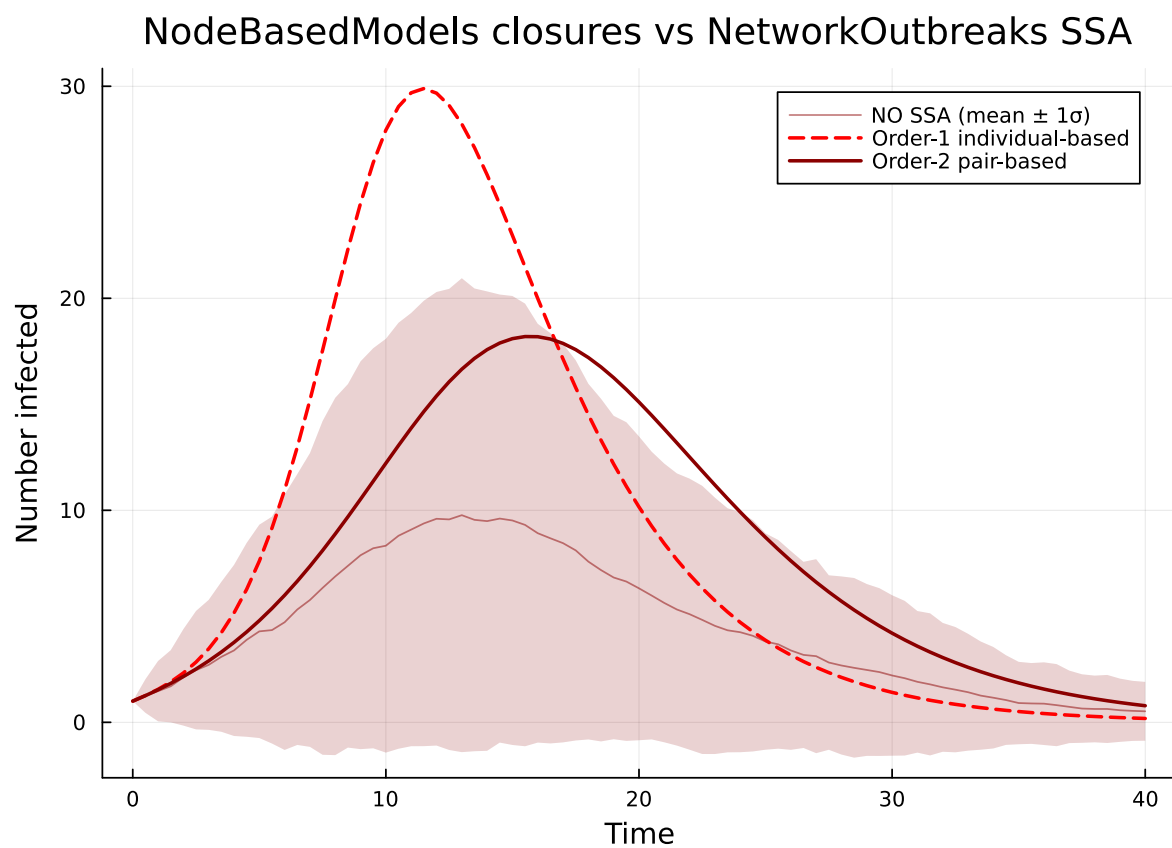


Figure 1: Figure 1: NetworkOutbreaks SSA ribbon (matched per-closure colors) over the order-1 and order-2 closures.